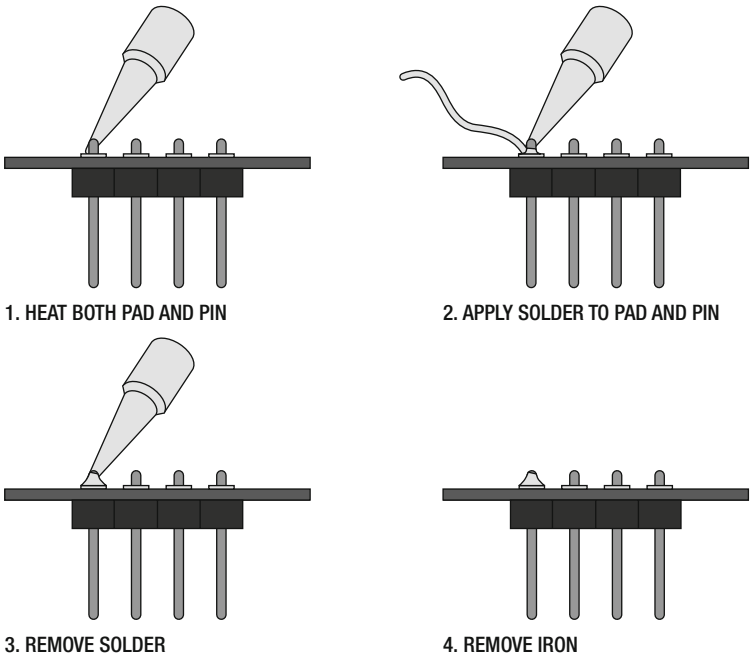

Figure 12-17. *Soldering*

Our example assumes that you want to solder pin headers to a printed circuit board, or PCB, although this process is fundamentally the same as soldering two wires together. Generally, we want to equally heat both parts with the tip of the iron and apply the solder to the two heated parts to be soldered and not directly to the soldering iron tip. We do this by putting the two parts together so that the metal bits are in contact, and then place the tip of the iron so that it touches both of these parts. In reality, we are using both the very tip and the side of the tip to make a good enough contact with both parts. In our example, we are soldering the short side of the pin header to the copper pad surrounding the hole that the pin was fed through on the circuit board.

When the parts are sufficiently heated, solder can be applied to the parts and it should flow quickly and smoothly when there is enough heat and everything is clean. Sometimes we will touch the solder to the tip of the iron to start the flow of molten solder, but we want to keep the solder on the parts to be soldered and not the iron. Remove the solder when a nice little mountain of solder has formed, but leave the iron heating things for another second or so. Once the solder has had a chance to flow over the whole pad, we can then remove the iron allowing the joint to cool naturally.

When the joint has been soldered correctly, you should have a clean and shiny little mound of solder that completely fills the copper pad and fully encompasses the pin. Sometimes when we don't add enough solder to the joint, we will end up with a pin that is only partially attached. This might work initially, but can create problems later when you least expect it. The simple fix is to simply heat up the joint and add more solder. However, if we add too much solder, we will end up with an enormous blob that might come into contact with other nearby pins or pads on the board. To fix this, we just need to heat up the solder again and shake or tap the board to remove the extra solder. Other times, instead of a shiny little mountain, we might end up with a solder joint that is dirty and dull looking. This is what's called a cold solder joint and happens when the tip is either not clean or not hot enough for the job.